

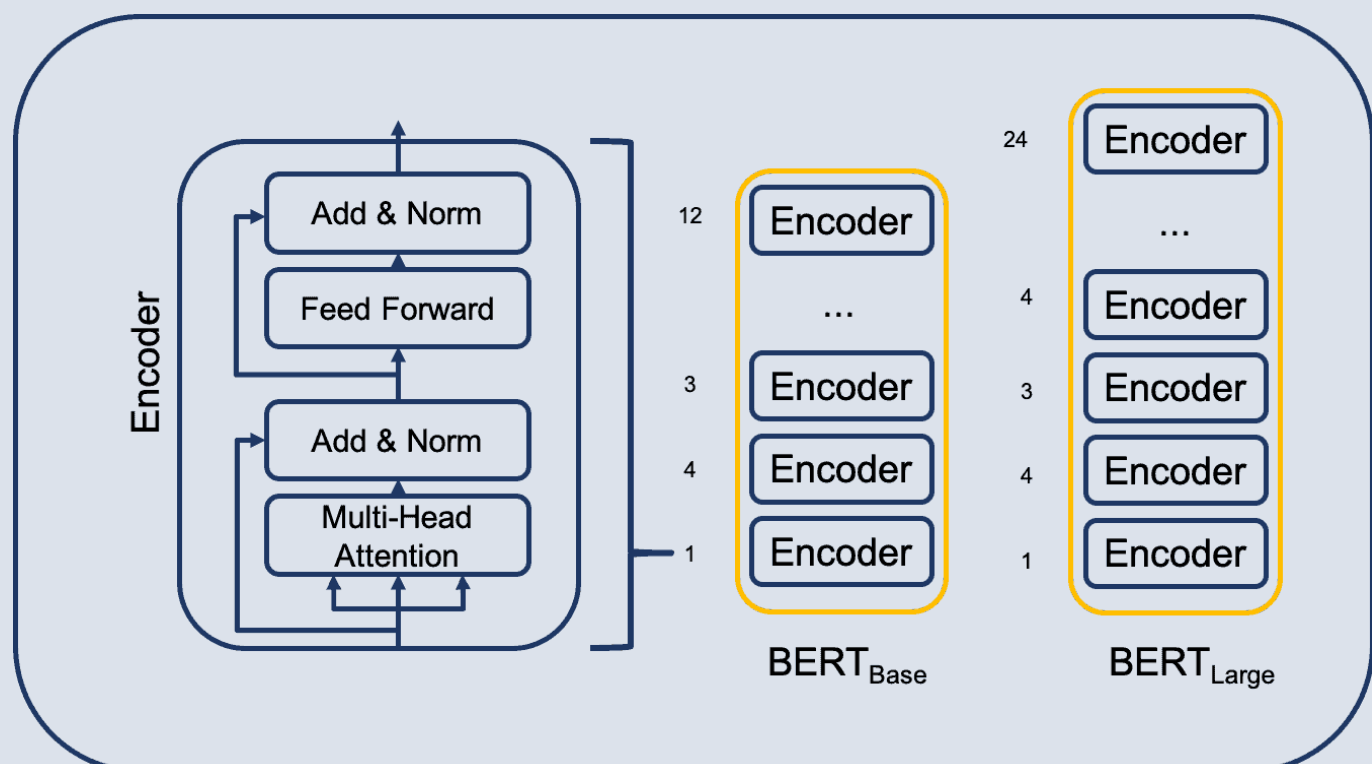
Deepfake Text Detection Systems for Arabic and English Using BERT and ELECTRA-Based Models



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CS4083 - NATURAL LANGUAGE PROCESSING & TEXT MINING

MODELS USED



- MODELS FOR DETECTING ARABIC: ARAELECTRA - ARABERT LARGE - ARABERT BASE
- MODELS FOR DETECTING ENGLISH: DISITBERT - ENGBERT - DEBERTA-V3-BASE

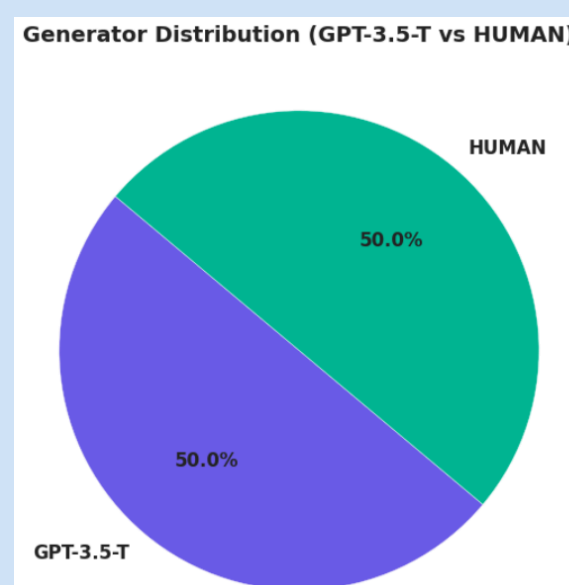
INTRODUCTION

AI-generated text, produced by tools like ChatGPT, Gemini, and Claude, is now widespread in many areas, making it increasingly difficult to distinguish from human-written content.



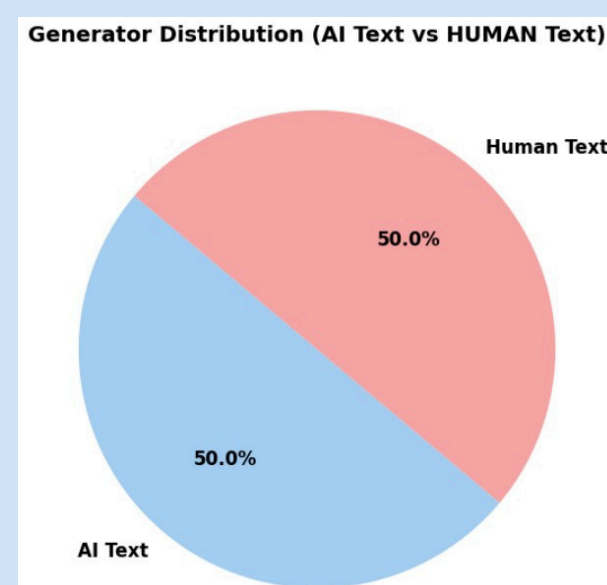
DATASETS

Arabic Dataset



ALHD

English Dataset

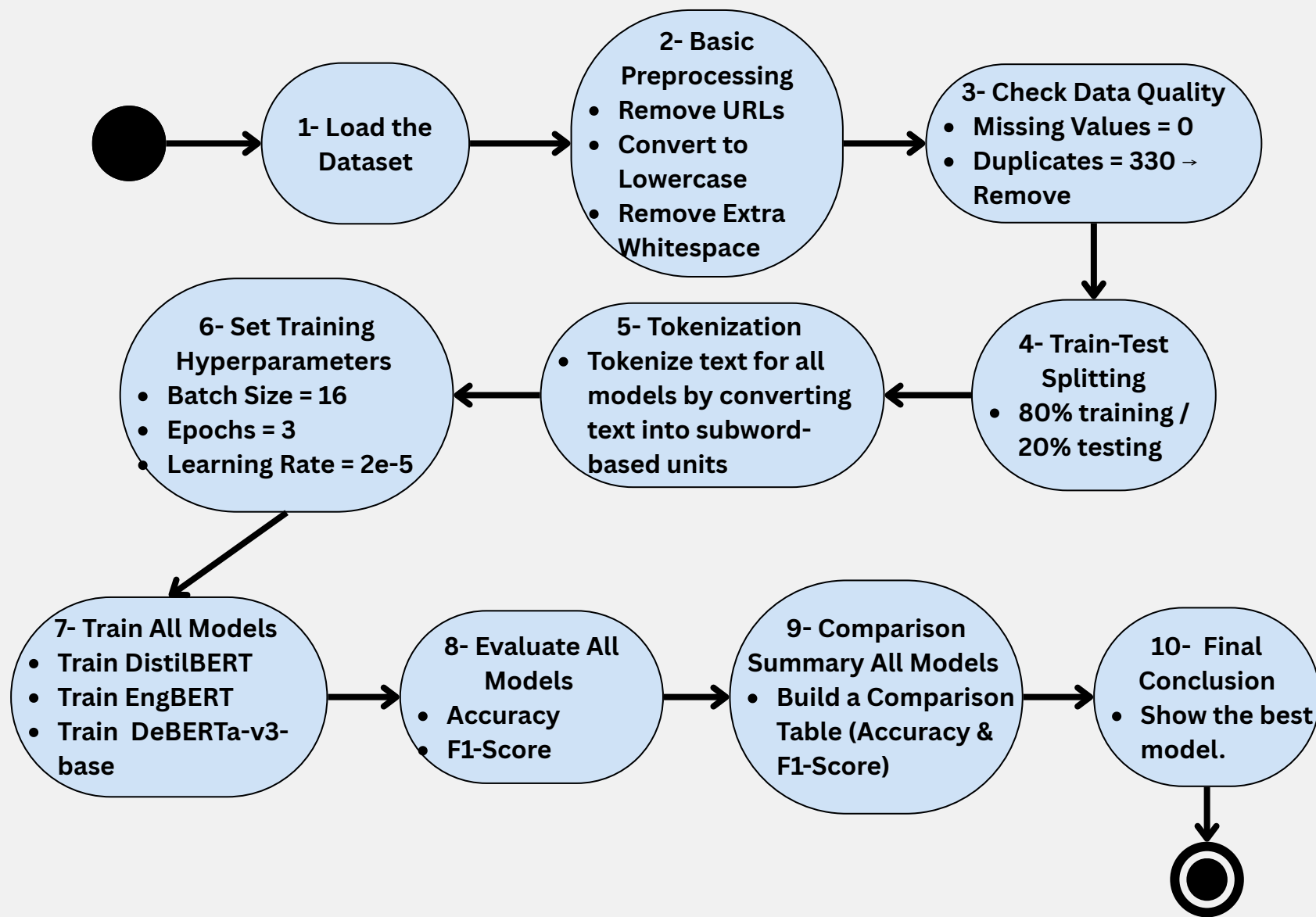


BALANCED_AI_HUMAN_PROMPTS

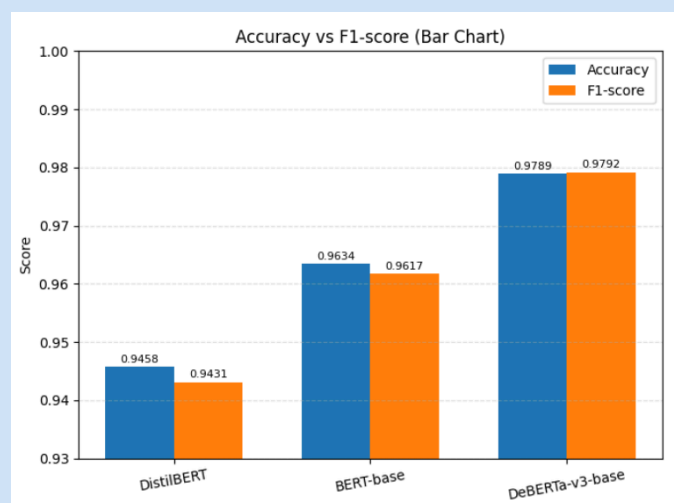
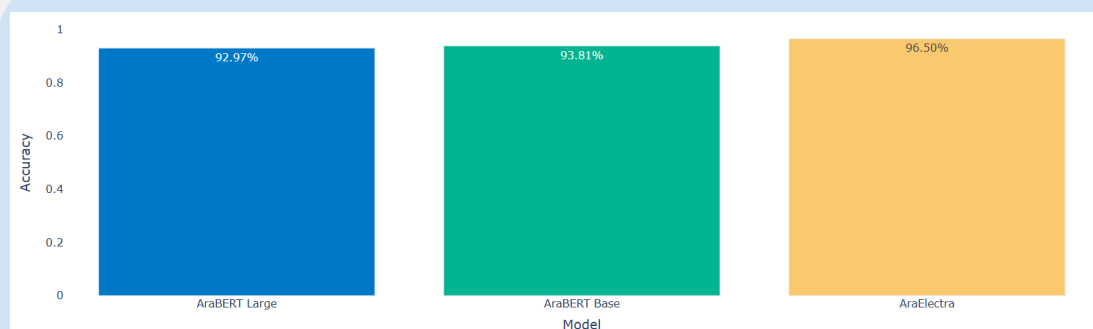
Preprocessing (Arabic)

Step	Description
Load Dataset & Basic Exploration	Already clean and no missing values
Splitting the Dataset	72% training - 20% testing - 8% validation
Tokenization of Text	- Text → input IDs and attention masks - Truncation: Long text is shortened to 128 tokens - Padding: All sequences are aligned to the same length - Normalization handled automatically by tokenizer
Creating a PyTorch Dataset	This prepares the data for batching during training.
Preparing Text for the Model	Converting each input into tensors

Preprocessing (English)



ARABIC/ ENGLISH MODELS RESULT



INPUT:

تعتمد الشركة إطلاق نظام ذكاء اصطناعي متقدم قادر على تحليل النصوص العربية بدقة عالية، وقد تم تدريب النموذج على ملايين الجمل، لكن تقارير أخرى تدعي أن النظام يتسبب في أخطاء أخلاقية غير مقصودة

OUTPUT:

Model	label	fake_probability
AraBERT-base	FAKE	0.9955
AraBERT-large	FAKE	0.9814
AraElectra	FAKE	0.9996